



Optimizing Promotional Strategy and Sales Performance Across Walmart Stores

GROUP 9

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Big Picture Problem



Why it matters?

a 1% increase in weekly revenue
chain-wide amounts to \$1-2 billion
in annual sales

How should Walmart US management adjust promotional and inventory strategies across store formats to improve weekly sales performance?

*A Data-Driven Analysis of Promotions,
Seasonality, and Store Performance*

Ask the Questions



1/ Which stores have grown or declined the most in weekly sales over the observed period (2010-2012)



2/ How stable and consistent are store sales trends over time?



3/ Do holiday weeks produce statistically different sales patterns than non-holiday weeks, and is this consistent across store types?



4/ To what extent do promotional markdowns increase weekly sales, after controlling seasonality and store factors?



5/ Among promotions, holidays, store attributes, and external factors (fuel price, CPI, unemployment...), which factors have the strongest impact on weekly sales



6/ Based on these findings and sales forecasts, how should management adjust markdown strategies and resource focus across store/department segments to maximize sales?

Data & Limitations

`features.csv`

`store.csv`

`test.csv`

`train.csv`





Descriptive Analytics





Store Performance Overview

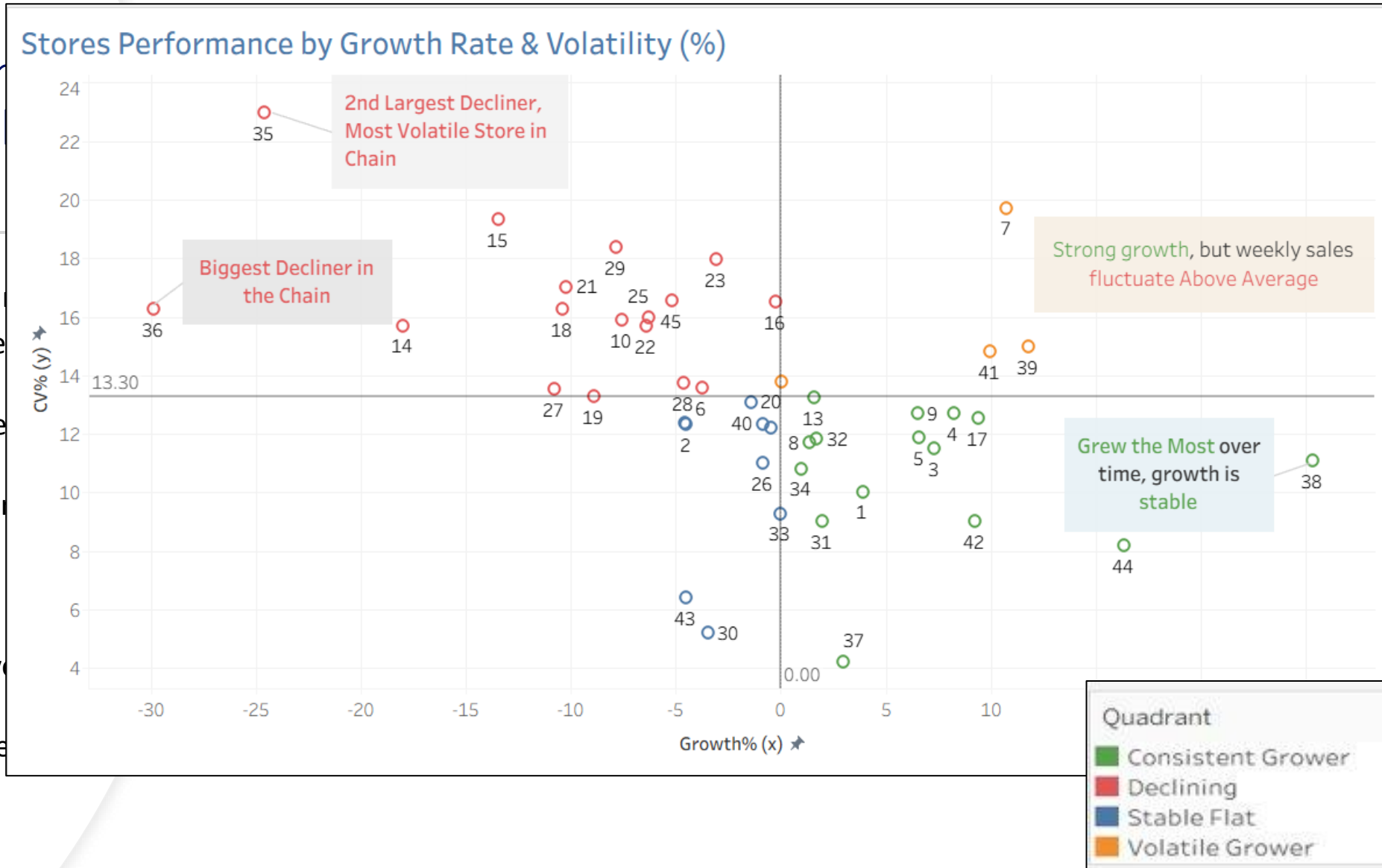
Sales levels vary significantly across 45 stores

Some stores consistently outperform others

No clear uniform trend across all stores

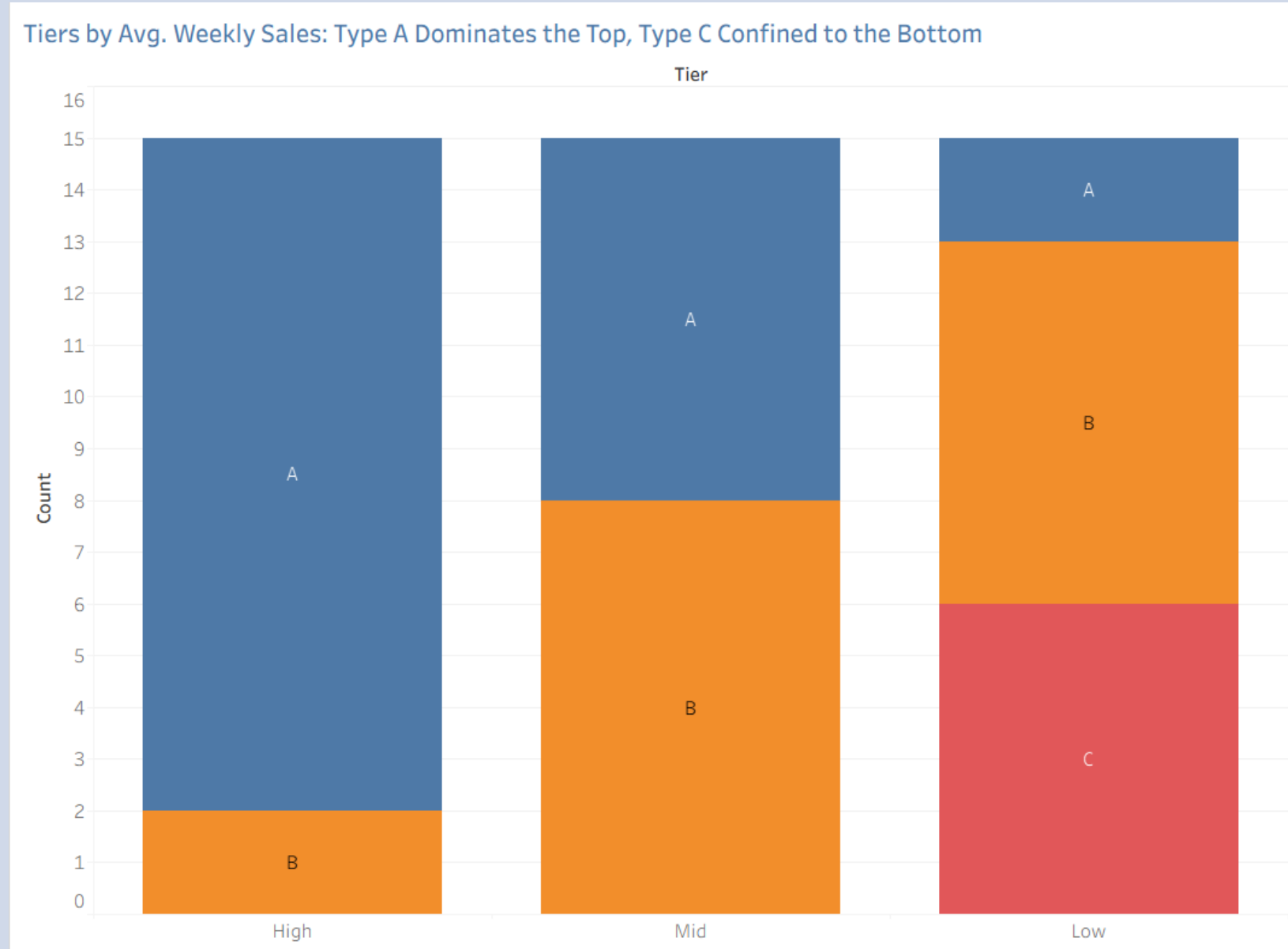
Indicates structural differences between locations

- Most stores are not growing
- Growth comes from only a few strong stores
- Store 38 is the best performer with strong and stable growth
- Some stores (like 7 and 39) are growing but unstable
- Many stores are declining or underperforming
- Overall, performance is uneven across stores
- This means we should not treat all stores the same



Growth Rate and Sales Volume Are Two Different Things

- Store 38, a Type C store, grew the most over time, but its average sales volume (\$386,000) is still less than 40% of what a typical Walmart store generates weekly.



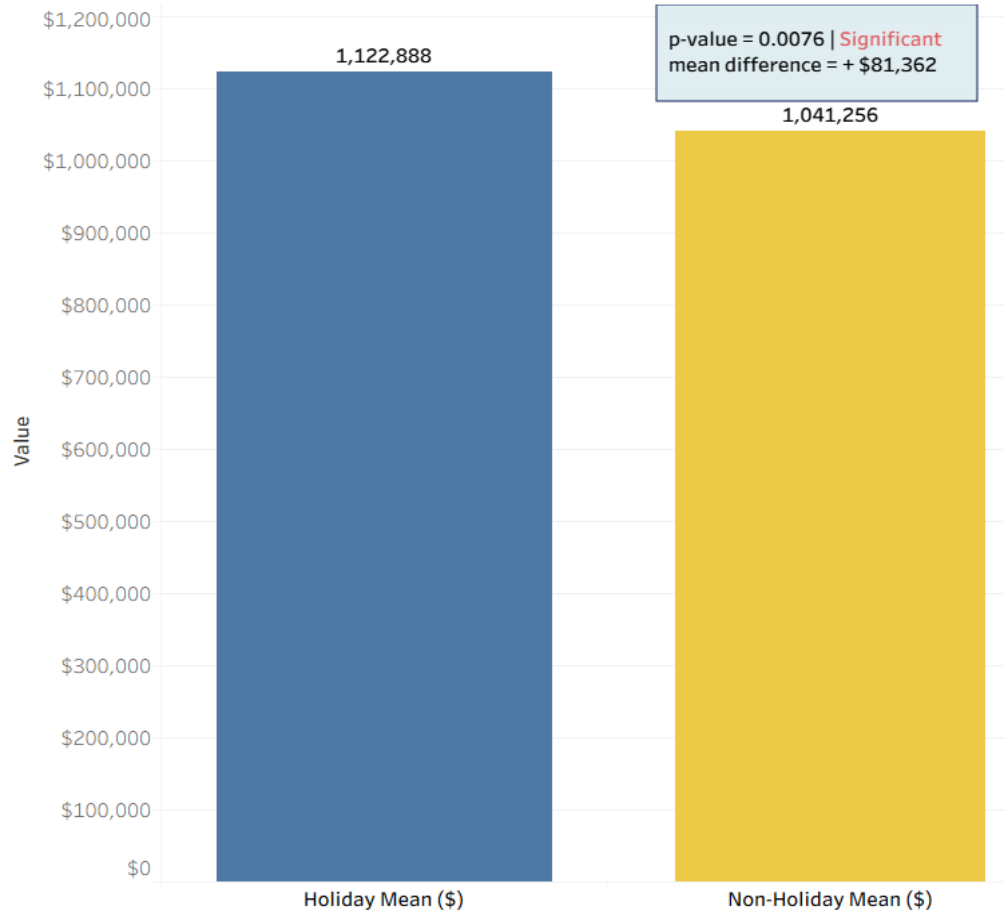
Diagnostic Analytics

- Why do weekly sales vary? And what can Walmart control to change it?

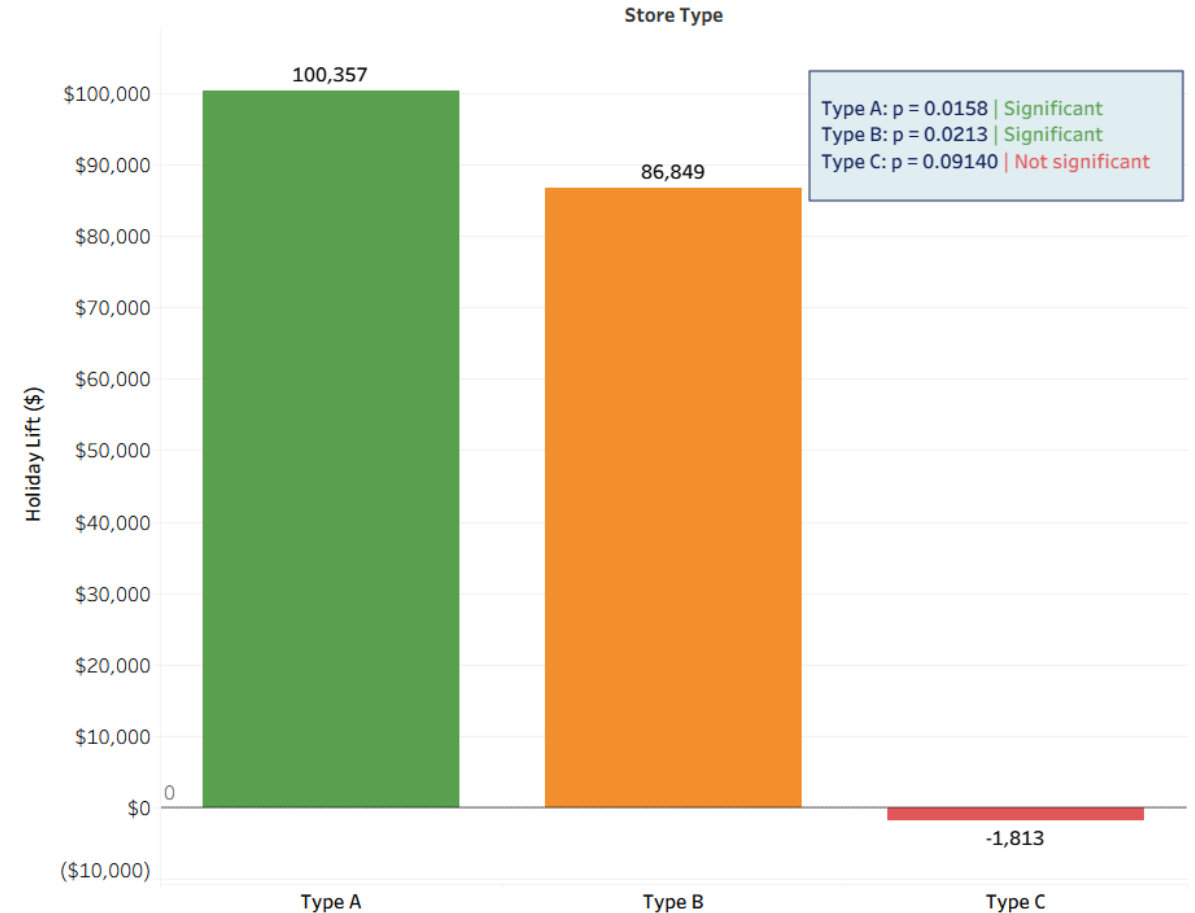


Diagnostic Finding 1: Holiday effect is REAL for weekly fluctuations in sales, but only significant in type A & B stores

Avg. Weekly Sales: Holiday vs. Non-Holiday

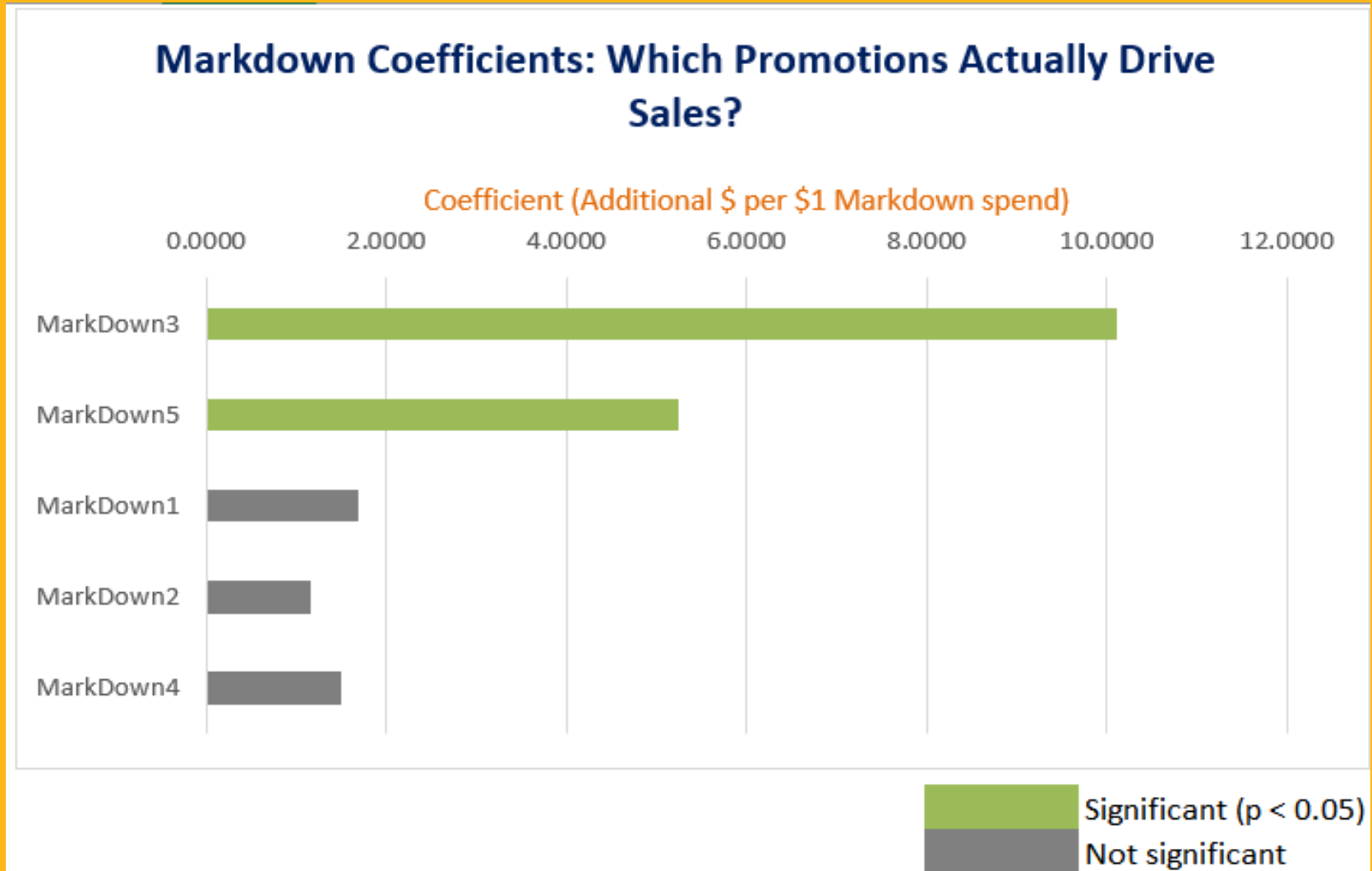


Holiday Lift in Avg. Sales per Store Types



Diagnostic finding 2a: Controlling for Store Size, Types, and Seasonality, only **MarkDown3** and **MarkDown5** show a measurable return.

- Controls held constant:
Holiday timing, Store
Size, Store Type
- OLS Regression, N =
2,295, $R^2 = 68.5\%$



Store Type	Markdown R ²	F-stat	p-value	Responds to Promotions?
Type A	16.40%	36.36	< 0.001	Yes
Type B	13.40%	22.15	< 0.001	Yes
Type C	2.30%	1.16	0.33	No

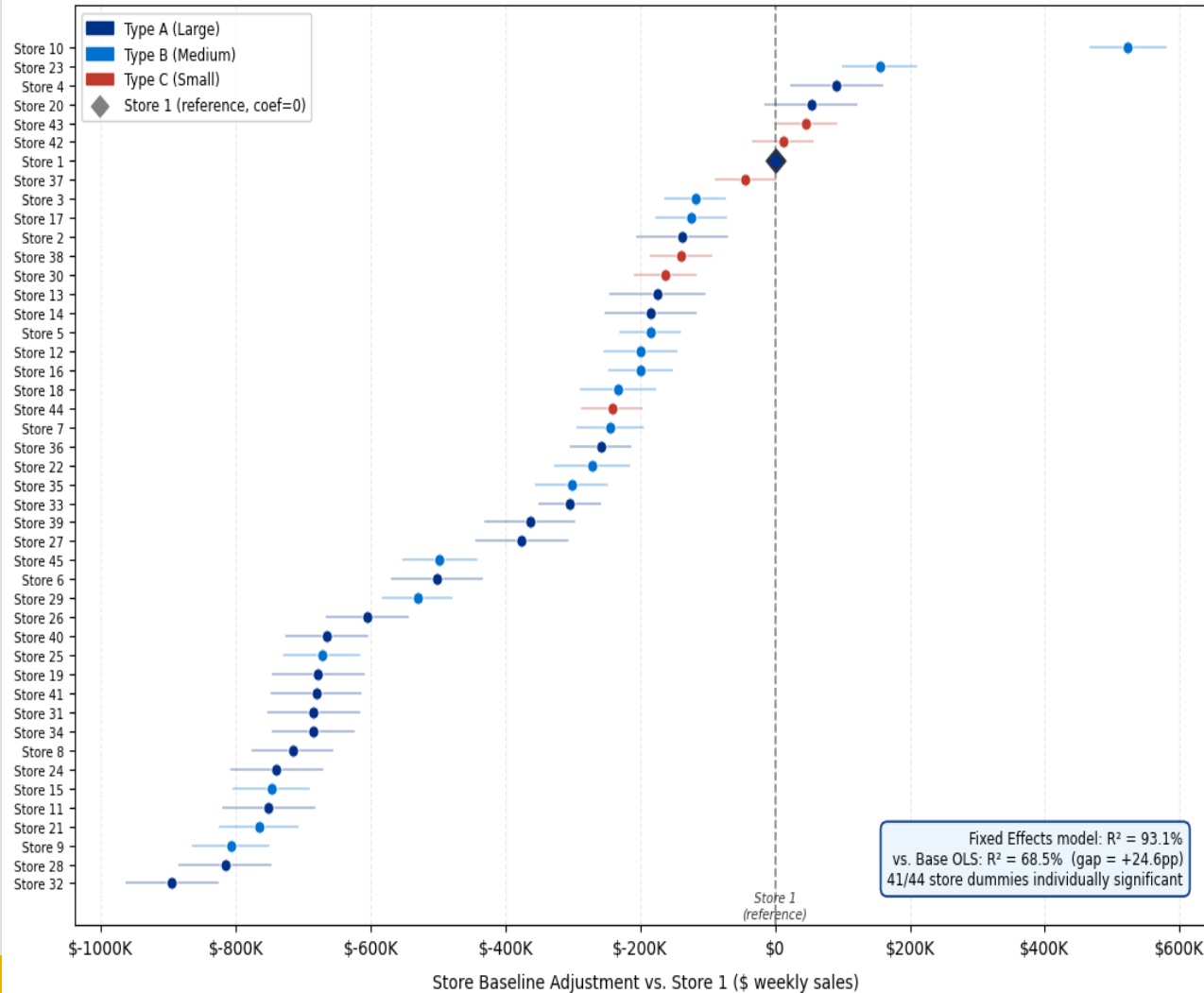
Running regression separately for each store type, Markdowns explain nothing in **Type C** stores.

Store	Avg Weekly Sales (\$)	Avg MD Spend/wk (\$)	Expected Lift/wk (\$)	ROI Ratio	Decision
30	436,336	2,534	-4,063	-1.60	CUT
37	525,872	1,818	627	0.34	PROTECT
38	428,259	2,157	13,489	6.25	PROTECT
42	578,400	1,840	24,905	13.53	PROTECT
43	626,253	1,799	7,366	4.09	PROTECT
44	326,650	1,481	6,593	4.45	PROTECT

Computing ROI for all 6 stores in type C, **store 30** expects both negative sales lift and ROI

Diagnostic Finding 2b: Type C stores do not respond to Markdowns, & store 30 is actively losing return

Individual Store Fixed Effects — Baseline Sales Premium / Discount
(OLS with 44 Store Dummies, N = 2,295 Strategy-A store-weeks)

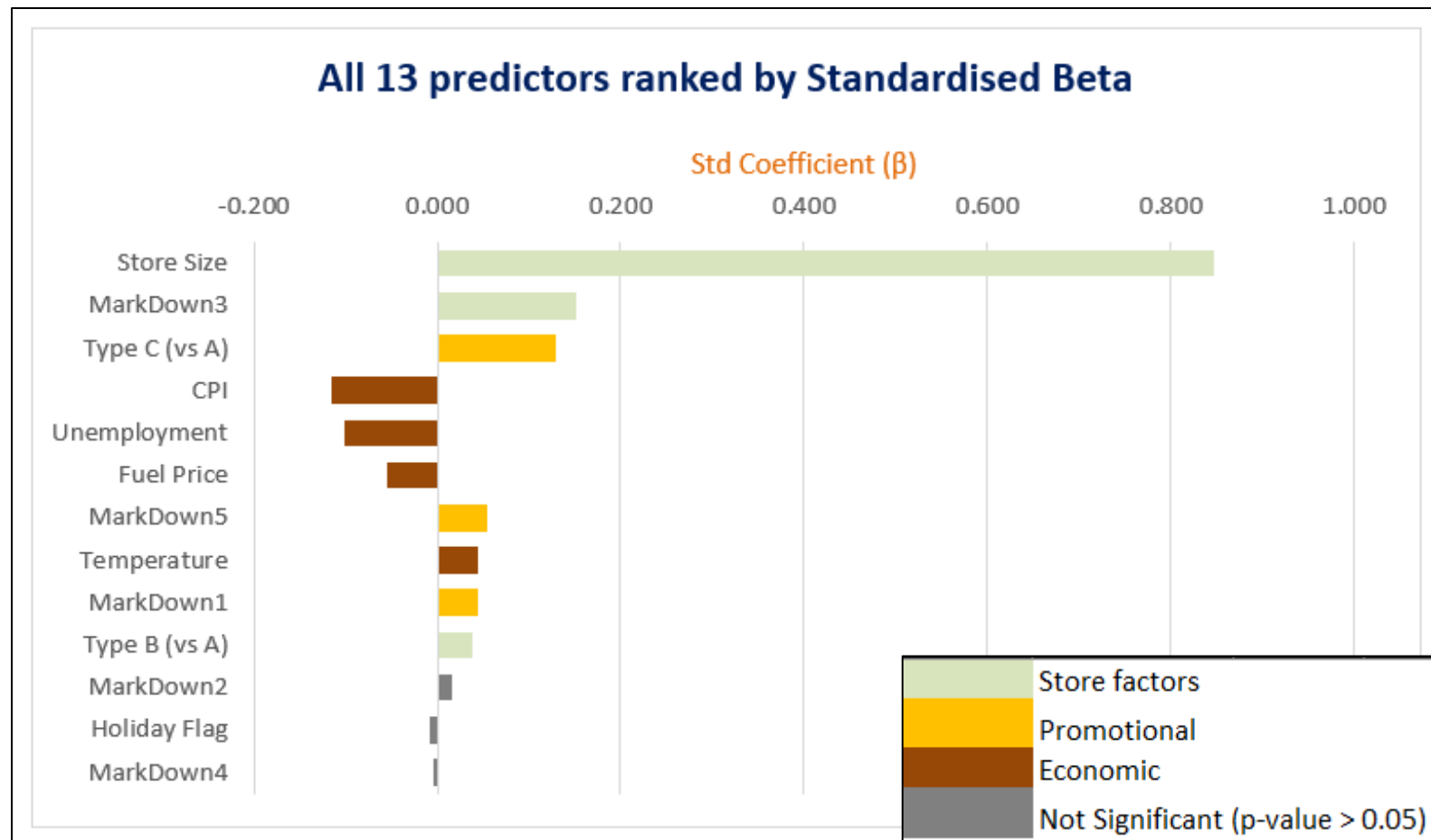


Comparison: Traditional Excel Toolpak OLS Regression v.s AI Fixed Effect Model

AI's Store-level fixed effects raise R^2 , BUT do not change which Markdowns matter.

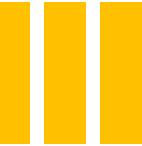
Diagnostic Finding 3

- Store size has the strongest relationship with weekly sales; 5x larger than any promotional factor. It sets the ceiling but cannot be changed in a budget cycle.
- Among factors Walmart can control, only **Markdown3** and **Markdown5** show a consistent, significant positive return across every model tested.



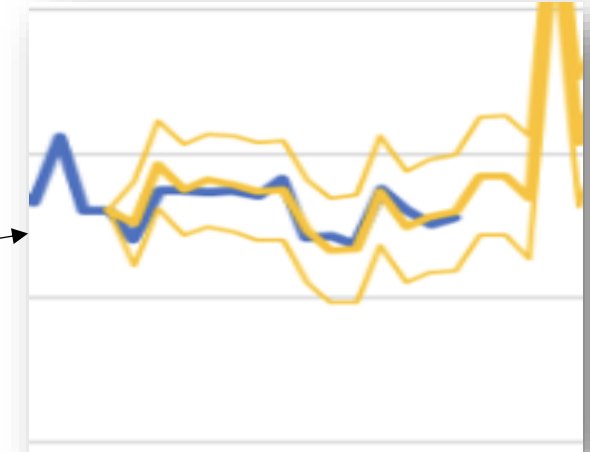
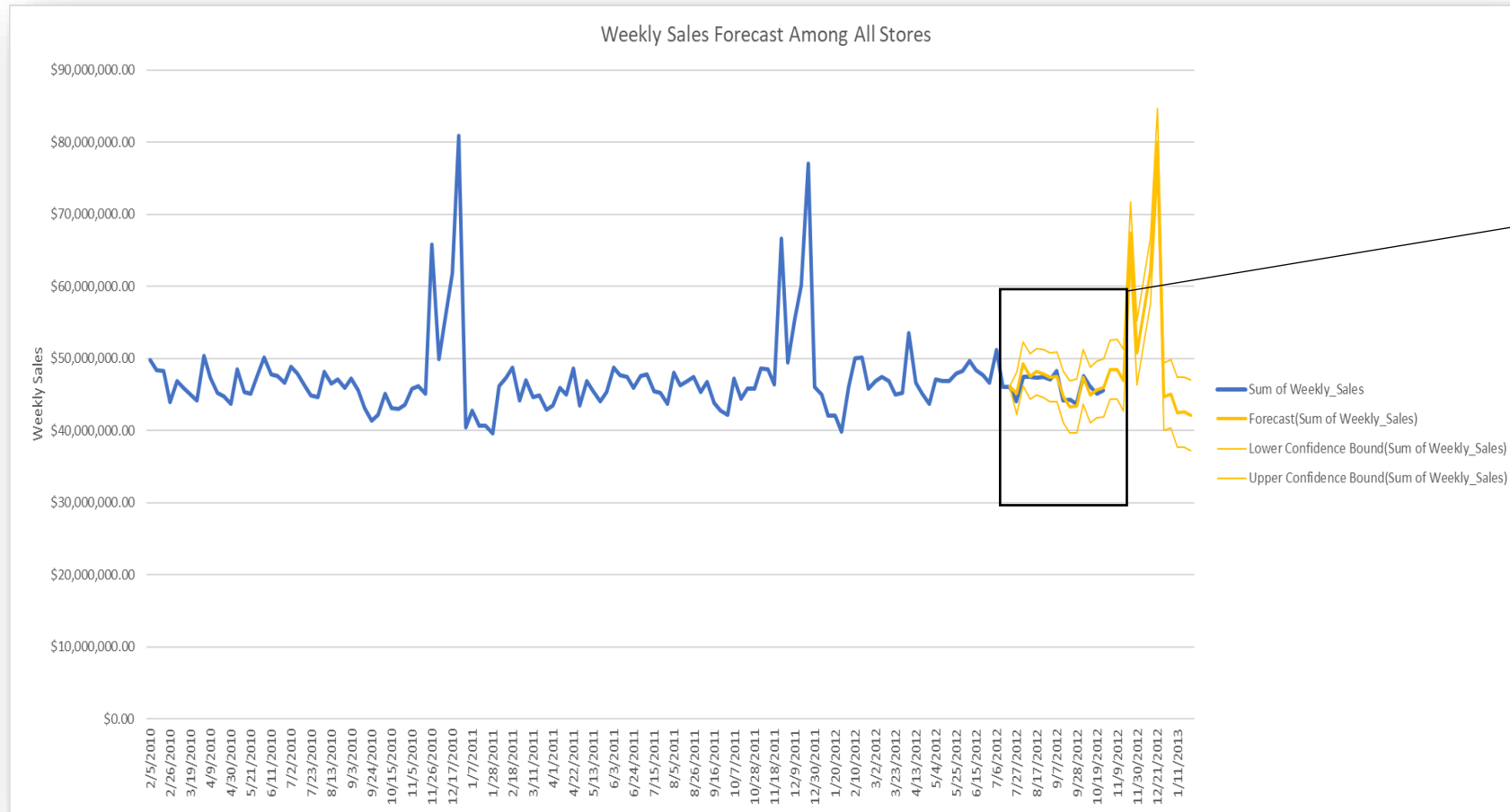


Predictive Analytics



Predictive Finding 1:

- Holdout Rate over 14 Periods = -0.352%
- Figured out that weekly sales forecast was consistent and stable over the period of 2010 - 2012



Week	Difference in Sales	% Change from Forecasted
7/27/2012	(\$1,104,839.36)	-2.505%
8/3/2012	(\$1,749,731.53)	-3.685%
8/10/2012	(\$102,852.46)	-0.217%
8/17/2012	(\$808,230.32)	-1.707%
8/24/2012	(\$456,667.28)	-0.962%
8/31/2012	(\$253,137.80)	-0.537%
9/7/2012	\$869,489.59	1.799%
9/14/2012	(\$441,870.07)	-0.999%
9/21/2012	\$1,071,136.44	2.415%
9/28/2012	\$329,438.75	0.753%
10/5/2012	\$157,070.52	0.330%
10/12/2012	\$1,170,259.96	2.537%
10/19/2012	(\$573,254.85)	-1.270%
10/26/2012	(\$400,594.25)	-0.880%
Totals	(\$163,841.62)	-0.352%

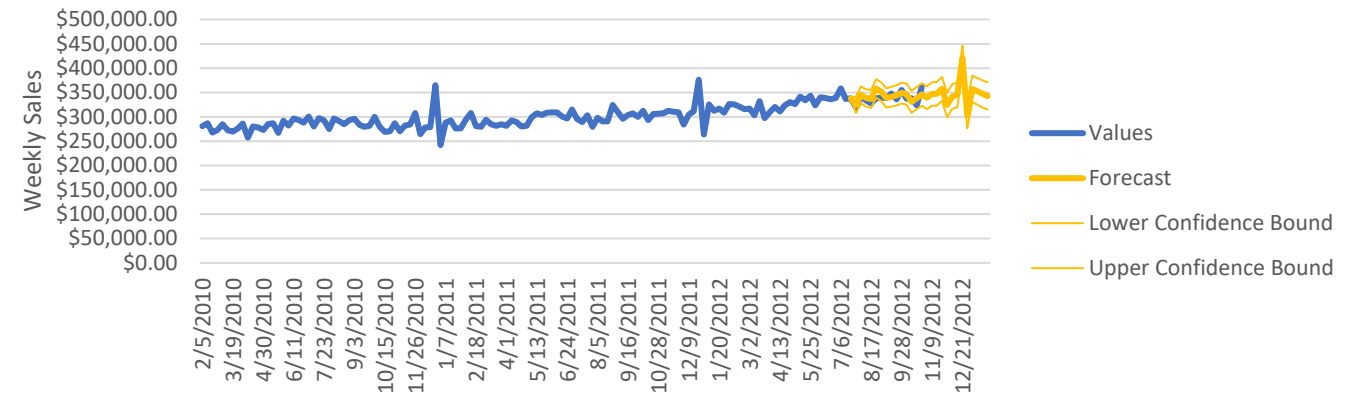
Predictive Finding 2:

- Held out 14 periods for each of the weekly sales forecasted for various stores based on growth and volatility
- Store 44 Holdout Ratio = -0.0115%
 - Strong, stable growth
- Store 35 Holdout Ratio = 0.002%
 - Negative, volatile growth
- Store 20 Holdout Ratio = -1.644%
 - Constant growth and volatility

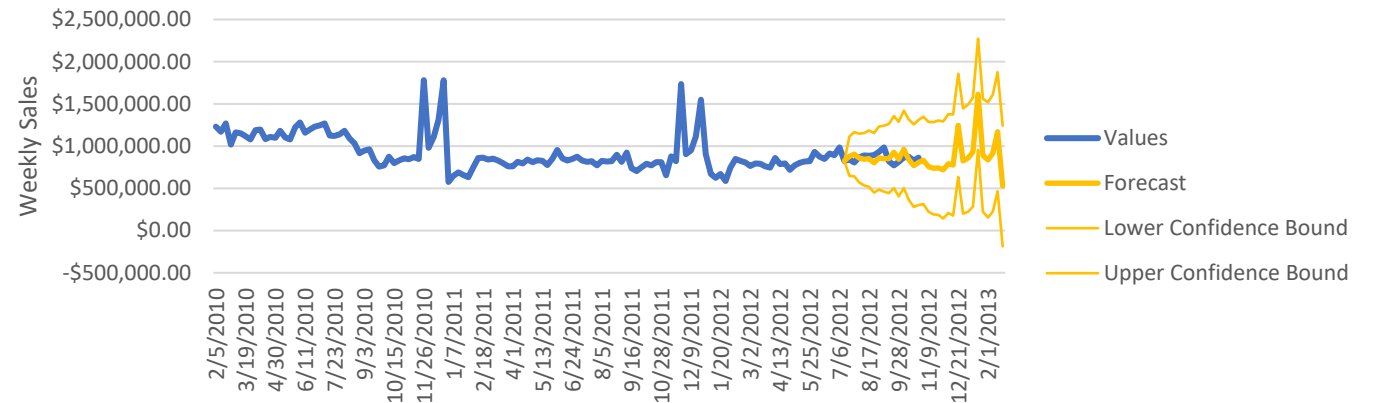
Conclusion:

- We effectively decided that store growth and volatility does not play an integral part into a stores forecasted weekly sales.
- We also conclude that we can forecast sales for all kinds of growth and volatility with enough confidence due to the holdout period.

Weekly Sales for Store 44



Weekly Sales for Store 35

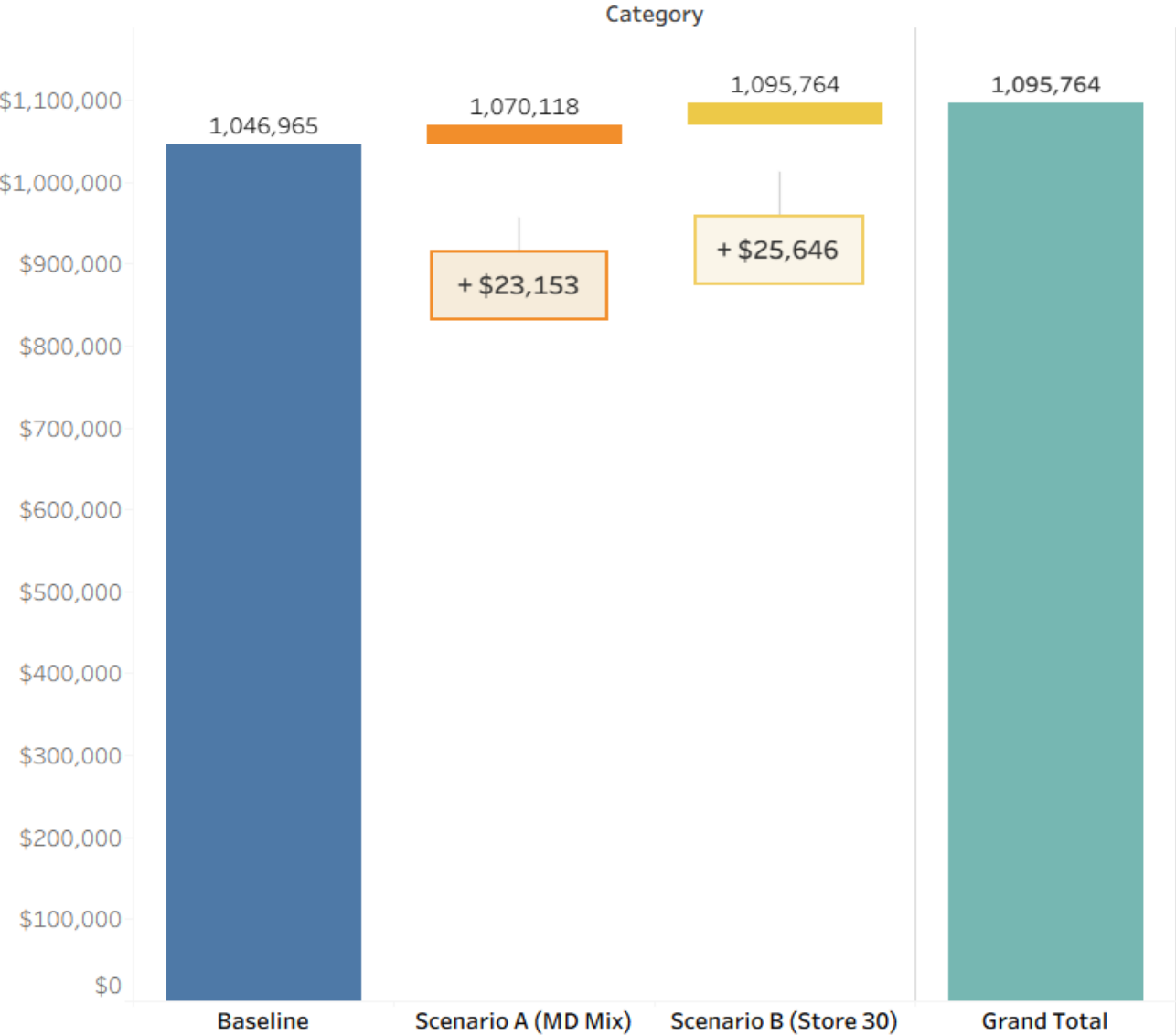


Weekly Sales for Store 20



Prescriptive Analytics: A total increase of \$48,799 per store/week if we act on recommendations

Weekly Revenue Increase (\$) by Recommendation



Recommendations

WHAT	Relocate 20% of ineffective markdown budget (MD1/MD2/MD4) into MD3 and MD5 in Type A and B stores
WHERE	Cut markdown spend from Store 30 (confirmed negative ROI) and redirect to MarkDown3 in Type A.
WHEN	Time both scenarios around holiday periods, where store responsiveness is confirmed and markdown and holiday effects compound



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your Time
& Attention
!